

①

Using ML, PL, NLP
Blockchain

Probability Distribution
Function for Data
Sec-22

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Lec-8 Uniform
Distribution

Lec-9 Log Normal
Distribution

Lec-10 Power Law
Distribution

① Continuous Uniform
Distribution

In probability theory
and statistics, the
continuous uniform distribution
or rectangular
distribution

such a distribution
describes an experiment
where there is an
ordinary outcome

→ Based on Continuous
Random Variable

The notation here is as:-

Notation :- $U(a, b)$

Parameter :- $-\infty < a < b < \infty$

If we talk about
PDF function

$$PDF = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

$$CDF = \begin{cases} 0 & \text{for } x < a \\ x-a & \text{for } x \in [a, b] \\ 1 & \text{for } x > b \end{cases}$$

$$\text{Mean} = \frac{1}{2}(a+b)$$

$$\text{Median} = \frac{1}{2}(a+b)$$

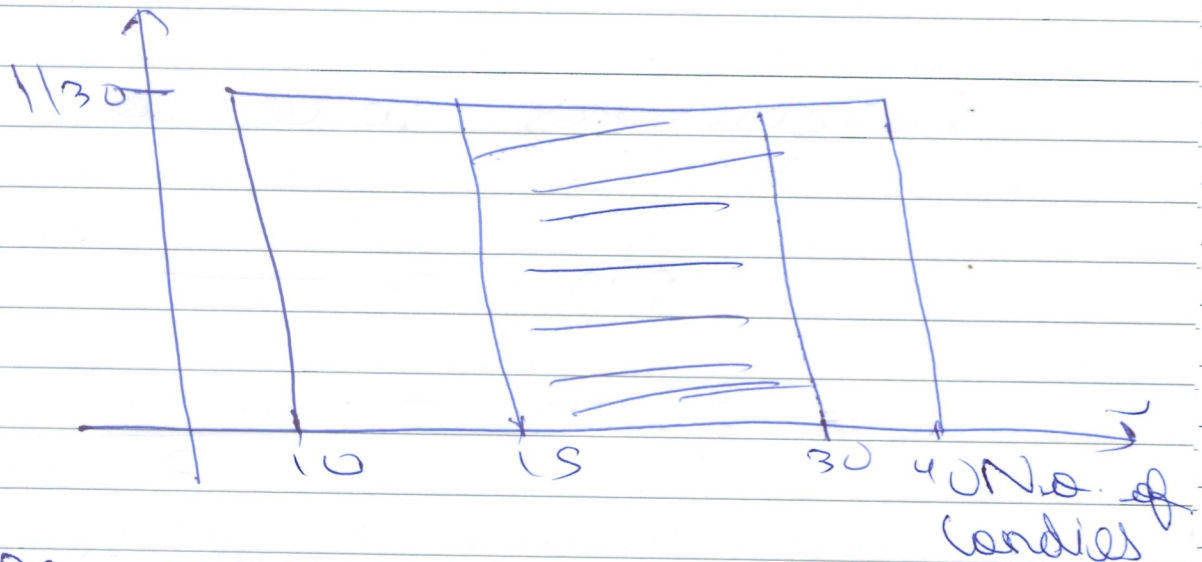
$$\text{Variance} = \frac{1}{12}(b-a)^2$$

Q.1) The number of candies sold daily at a shop is distributed uniformly with a range of 40 candies & mean of 10 candies.

Sol

(1) Probability of daily sales the shop being 15 to 30

Soln



$$x_1 = 15 \quad x_2 = 30$$

$$P(15 \leq x \leq 30) = (x_2 - x_1) \times \frac{1}{b-a}$$

$$= (30 - 15) \times \frac{1}{30}$$

$$= 0.5 \text{ Ans}$$

$$P(x \geq 20) = (40 - 20) \times \frac{1}{30}$$

$$= 66\%$$

The samples will be like :-

Rolling a die (Fair die)
 $\{1, 2, 3, 4, 5, 6\}$

$$\frac{1}{n} \Rightarrow n = b - a + 1 \\ = 6 - 1 + 1 = 6$$

Notation = $u(a, b)$

Parameters are a, b where
 $b \geq a$

$$PMF = \frac{1}{n}$$

$$\text{mean} = \frac{a+b}{2}$$

$$\text{median} = \frac{a+b}{2}$$

Lec - 9

Log Normal Distribution

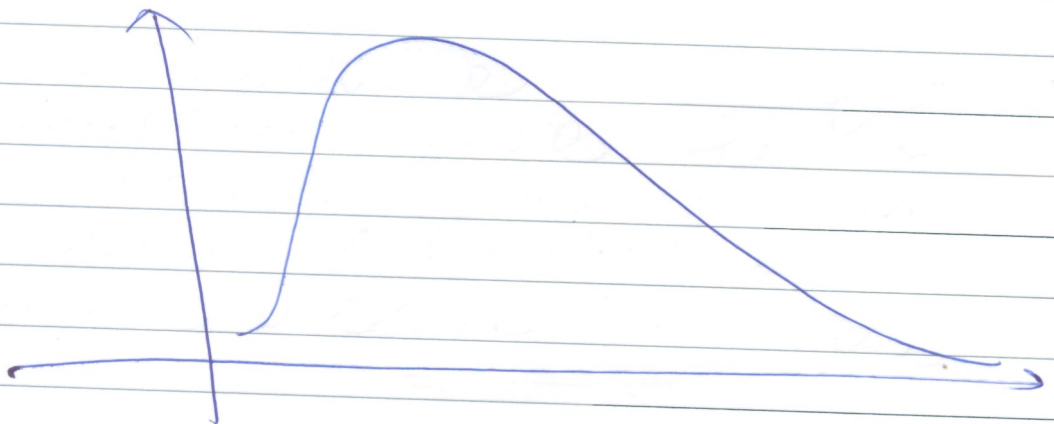
- 1. In the probability theory, a log normal distribution is a continuous probability distribution of a random variable whose logarithm is normally distributed.
- 2. If the random variable X is log normally distributed, then $Y = \ln(X)$ has a normal distribution.
- 3. If Y has a normal distribution then exponential function of Y , $X = \exp(Y)$ has log normal distribution.

 X (Random Variable) $X = \text{Log Normal Distribution}(\mu, \sigma)$

The mean is constant
but the standard
deviation is changing.

Hence Log - Normal
Distribution is a right
skewed distribution.

The graph can be plotted as



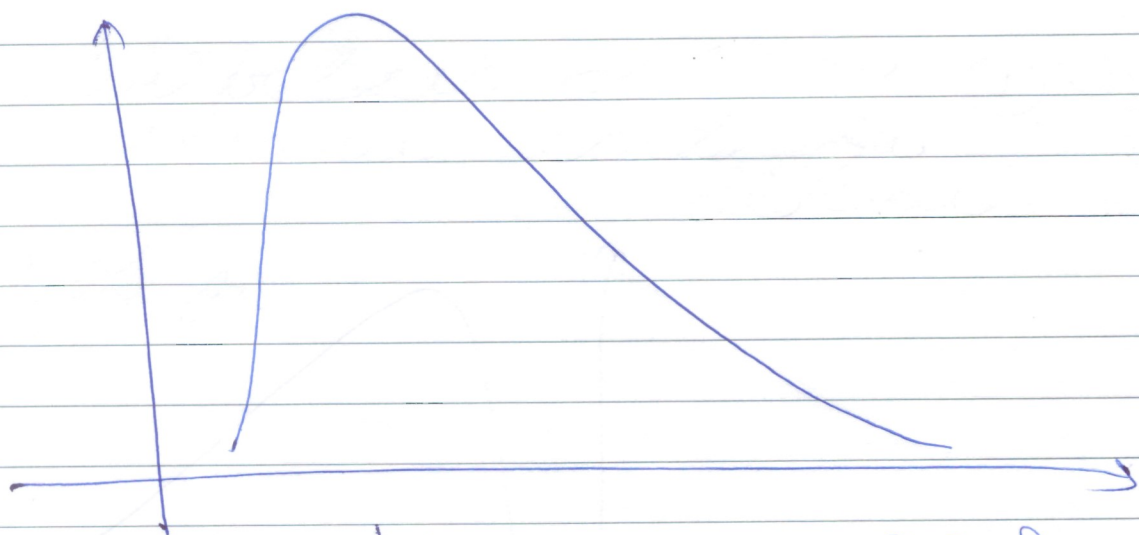
$y = \ln(x) \Rightarrow$ Normal
distribution



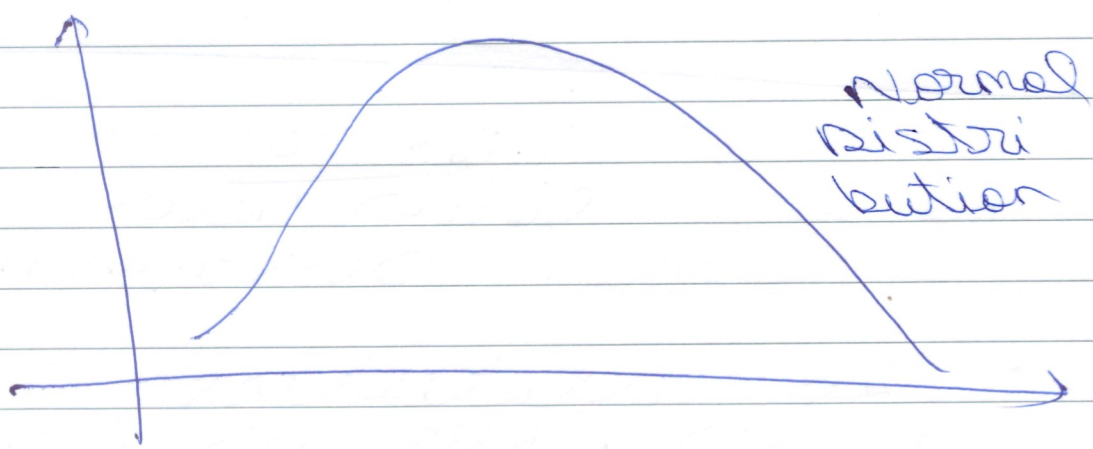
Normal by log (x)

$x \approx \exp(y) \Rightarrow$ Log
normally
distributed

7

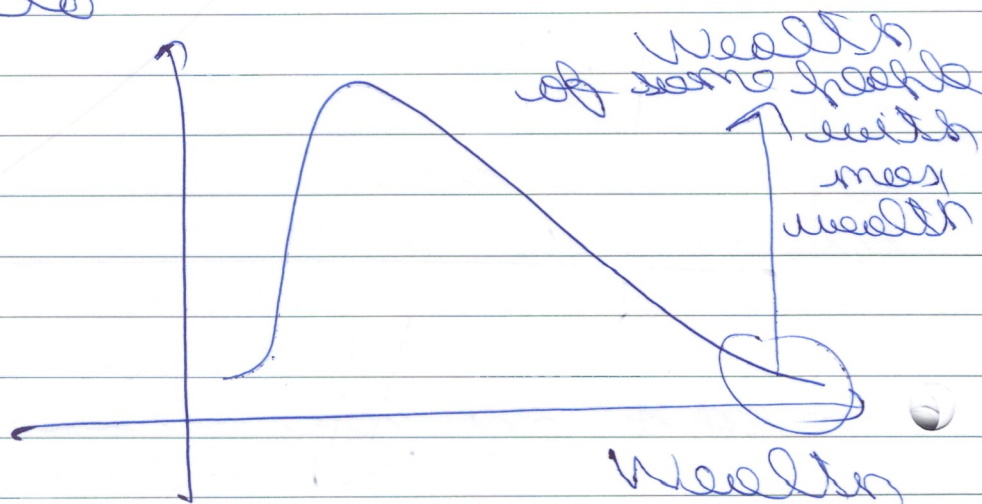


if we apply
 $\ln(x)$, we
get gaussian
distribution



For Normal distribution
we apply $\ln(x)$

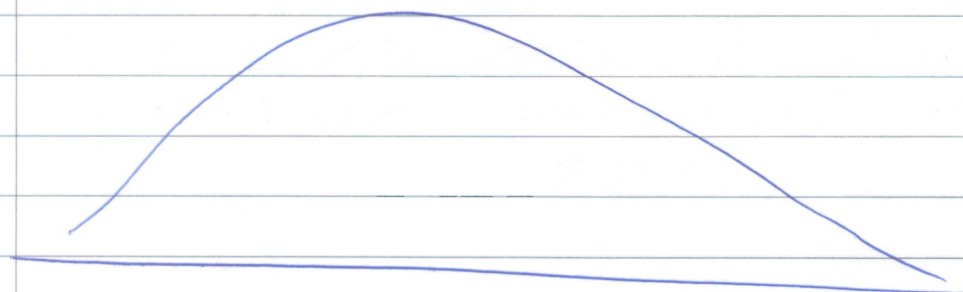
Ex: (1) Distribution of the whole world



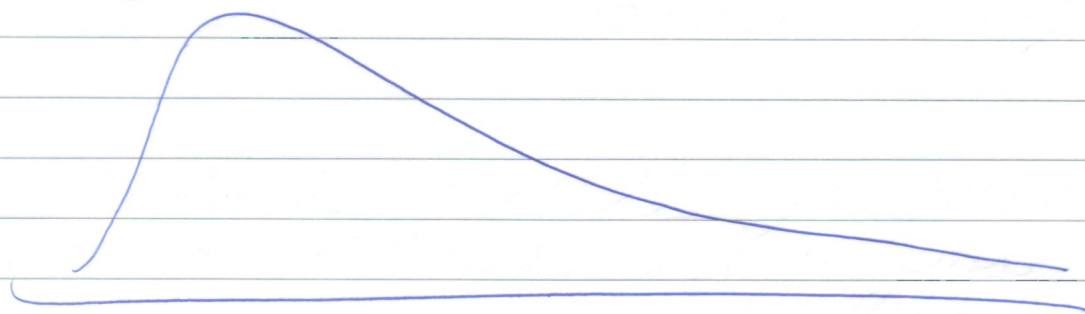
Ex - 10
Power Law Distribution

In statistics, the power law is a function or relationship (or distribution) over which the change in one or more quantities or variables is proportional to the change in another. It is distinct from the exponential distribution of these quantities.

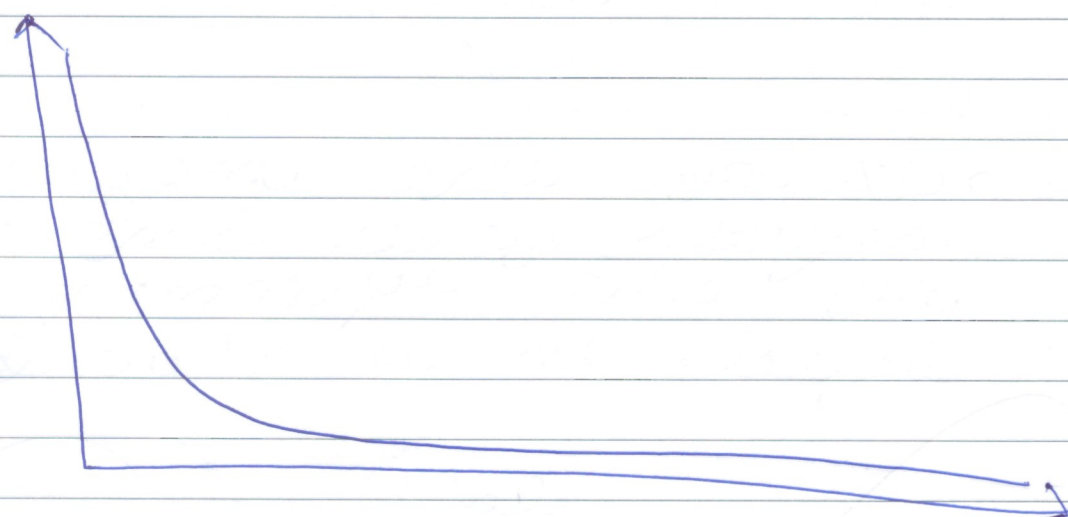
Normal Distribution



Log Normal Distribution:



Power Law Distribution



As one quantity increases, the other also increases.

Ex ① 2m TPL.

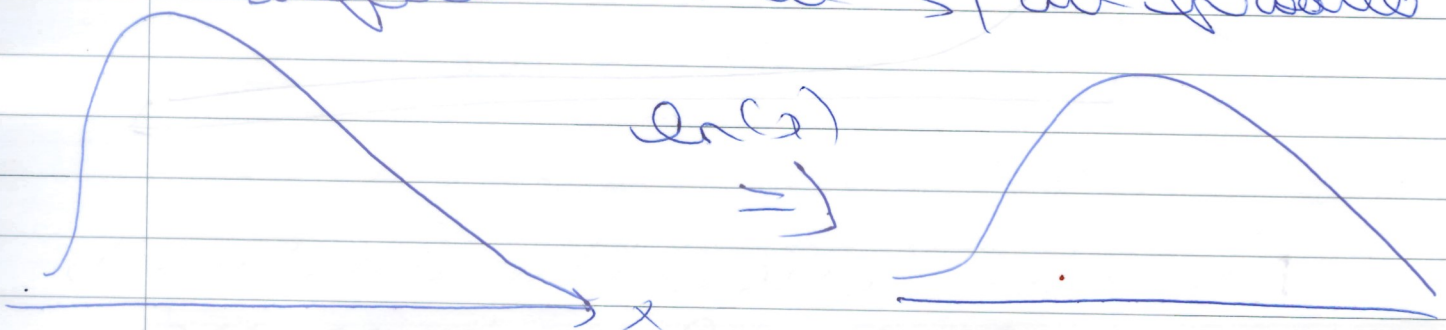
20% of team is responsible
for winning 80% of
the market

② 80% of wealth is
with 20% of the
population

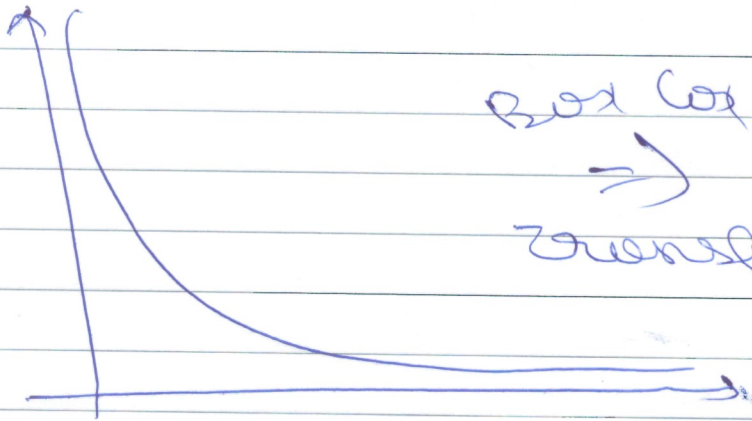
③ 80% of total
wealth is with 20% of
nation

④ 75% of words in
most languages

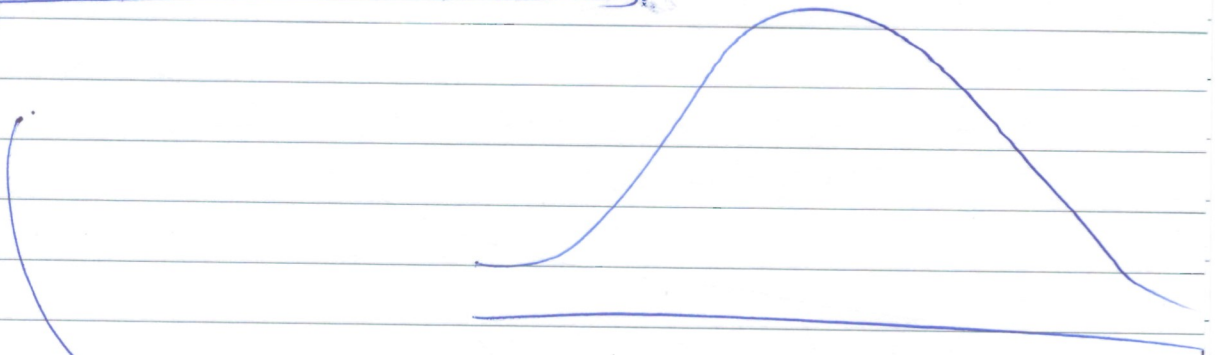
⑤ 20% of the major
defects gives 80%
of the upcoming
defects in a product



⇒ skewed data distribution



Box Cox
⇒
transform



⇒ This is
specifically
normal distribution